MySQL Scalar and Aggregate Functions

In MySQL, functions are categorized based on how many rows they process to produce a result.

Scalar Functions (which include System Information functions) operate on individual rows or system states.

Aggregate Functions operate on a set of values to return a single summary result.

1. Aggregate Functions

Aggregate functions perform a calculation on a set of values (a column of data) and return a single value. They are almost always used with the `GROUP BY` clause.

Function	Description	Example
`COUNT()`	Returns the number of rows that match a specific criterion.	`SELECT COUNT(id) FROM orders;`
`SUM()`	Returns the total sum of a numeric column.	`SELECT SUM(salary) FROM employees;`
`AVG()`	Returns the average value of a numeric column.	`SELECT AVG(price) FROM products;`
`MIN()`	Returns the smallest value in a column.	`SELECT MIN(age) FROM students;`
`MAX()`	Returns the largest value in a column.	`SELECT MAX(score) FROM results;`
`GROUP_CONCAT()`	Concatenates values from a group into a single string.	`SELECT GROUP_CONCAT(name) FROM users;`

Finding the total sales and average price per category :

```
SELECT category,
    COUNT(*) AS total_items,
    SUM(sales) AS total_revenue,
    AVG(price) AS avg_price
FROM products
GROUP BY category;
```

2. System Information Functions

These are a type of **Scalar Function**. Instead of processing table data, they return information about the database state, the current user, or the server version.

Function	Description	Example Output
DATABASE()	Returns the name of the current default database.	"ecommerce_db"
USER()	Returns the current MySQL user and host name.	"admin@localhost"
VERSION()	Returns the version of the MySQL server.	"8.0.32"
CONNECTION_ID()	Returns the unique connection ID for the current session.	142
LAST_INSERT_ID()	Returns the last AUTO_INCREMENT value generated.	1025
CURRENT_USER()	Returns the authenticated user name.	"root@%"
BENCHMARK(n, exp)	Executes an expression n times (used for testing speed).	0 (shows time taken)

```
SELECT
    DATABASE() AS current_db,
    USER() AS logged_in_as,
    VERSION() AS mysql_version;
```

3. Other Common Scalar Functions

While you specifically asked for System functions, most MySQL scalar functions fall into these sub-categories:

- **String Functions:** UPPER(), LOWER(), CONCAT(), LENGTH().
- **Numeric Functions:** ROUND(), ABS(), CEIL().
- **Date/Time Functions:** NOW(), CURDATE(), DATEDIFF().

Comparison Table

Feature	Aggregate Functions	System/Scalar Functions
Input	Multiple rows (a set)	Single value or system state
Output	One single value for the whole set	One value per row/call
Context	Data analysis/Reports	Data formatting/Auditing
Commonly used with	GROUP BY	SELECT, WHERE, INSERT